

## Some applications

**Intelligent prosthetics.** Prosthetic applications are the most powerful driver of research in BCI. There are millions of people who cannot see or hear or have lost a body part. BCI infuses intelligence into the otherwise well-designed prosthetic devices to bring them in sync with the mind. For example, the prosthetic arm should hold the object just as the user thinks about moving the prosthetic arm.

**Mind reading.** Non-invasive BCI has much higher potential. Imagine the power of being able to read a person's mind. This would find tremendous applications, all of which are hard to imagine now. However, as evident, such mind reading schemes will create ethical issues as well. As the intellectual ability of the person will be threatened, access to such mind-reading devices has to be very limited.

**Download skills the 'Matrix' way.** All our abilities are possible because of the information stored in the brain. The brain does the processing, refines information and stores the same. This storing mechanism, once understood, will make it possible to 'write' information to the brain—the way Neo and Trinity (characters in *Matrix* movie) downloaded programs for skill-development. This will open up tremendous opportunity for enhancing one's skills.

**Humans getting immortal.** With fast paced development of BCI, in another 20 to 30 years, the distinction between man and machine (mortal and immortal) will blur. In other words, humans and android will become synonymous. Copying mind is already on the verge of reality. The next step would be to store the copied mind to a semiconductor memory.

upon the frequency, amplitude components and source of the channel from which the spike-train was recorded. Each such signature is then associated to a particular physiological process. Several such techniques are worked upon to compress and process neural data.

**Neural signal classification.** Associating the spike trains to a particular physiological activity is a daunting task with ample chances of making mistakes. Critical thinking is required for the right match of the correct signal source, its generated neural signal and the intended physiological behaviour. Cross-correlation analysis is used to reveal temporal and spatial relationships between multiple spike trains in a neuronal circuit, in which two neurons are synaptically connected or two neurons are independent but receive a common input. Researchers are using different mathematical techniques of pattern matching, etc to classify neural signals and associate them with the particular physiological phenomena.

**Prosthetic device.** Prosthetic devices surrogate human body parts to restore or enhance the functionality of the organ. Ideally, these look, perform and feel like a natural limb. As shown in the block diagram of a BCI system, a prosthetic device is the final recipient

of neural signal information to perform its operation and the neural processor is usually inbuilt in the prosthetic device. Examples of existing prosthetic devices are pacemakers, bionic eyes, ears, knees, feet, legs, arms and so on.

The prosthetic limb receives the command signal to engage itself for a particular task. For example, when a person (with a prosthetic arm) wants to hold certain object, he thinks of performing this task. Neuron signals in the brain are activated accordingly and understood by the neural processor. The processor then commands the device (prosthetic arm) to perform the needful, i.e., hold the object of interest.

There are two major approaches to develop the prosthetic devices: rehabilitate the patient for his lost/damaged body part or enhance the ability of the body part (beyond normal human abilities). The devices are categorised on the basis of their functionality—whether they augment muscular actions or that of the nervous system. Because each patient is unique, his prosthetic device has to be unique too. Research is underway on material selection for prosthetic devices to add more intelligence, increase reliability and so on. Advanced materials is a major research area for implementing prosthetics. Right toughness, light weight, durable and proper

bending strength are the ideal properties of a prosthetic device. Thanks to nanotechnology, realising the ideal prosthetic material looks within reach.

## Epilogue

BCI has received massive global attention. Interestingly, researchers are divided over two approaches to BCI—invasive and non-invasive—based on how the system is connected to the brain. The invasive technique would require a surgery to implant the probes into the brain, while the non-invasive technique would require a head-mounted device to be fixed over the skull. Invasive technique is popular in European and Japanese countries, while the other is more popular in American institutes/research centres.

In any case, it is only a matter of time before we stand face to face with this ultimate invention. Synergy of subjects from different fields would pave the way for achieving this goal. Researchers in this field have to unravel ambiguities of human emotions and feelings—ethics, good/bad, etc—to develop intelligent, thinking machines.

RGP Prosthetic Research Center, San Diego (USA), John Hopkins University, California Institute of Technology, Northwestern University, Vanderbilt University, University of Michigan, University of California (Irvine), University of Utah, University of Lausanne (Switzerland), IMEC (Belgium) are some of the renowned American and European institutes for prosthetics. National Brain Research Center in Gurgaon (Haryana) is doing challenging studies in the areas of neurology, neural computation, etc. But Indian research institutes need to move much faster to carve their niche in this important field. Fellowships and lucrative incentives to attract bright fellows into this fantastic interdisciplinary area is the need of the hour. ●

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